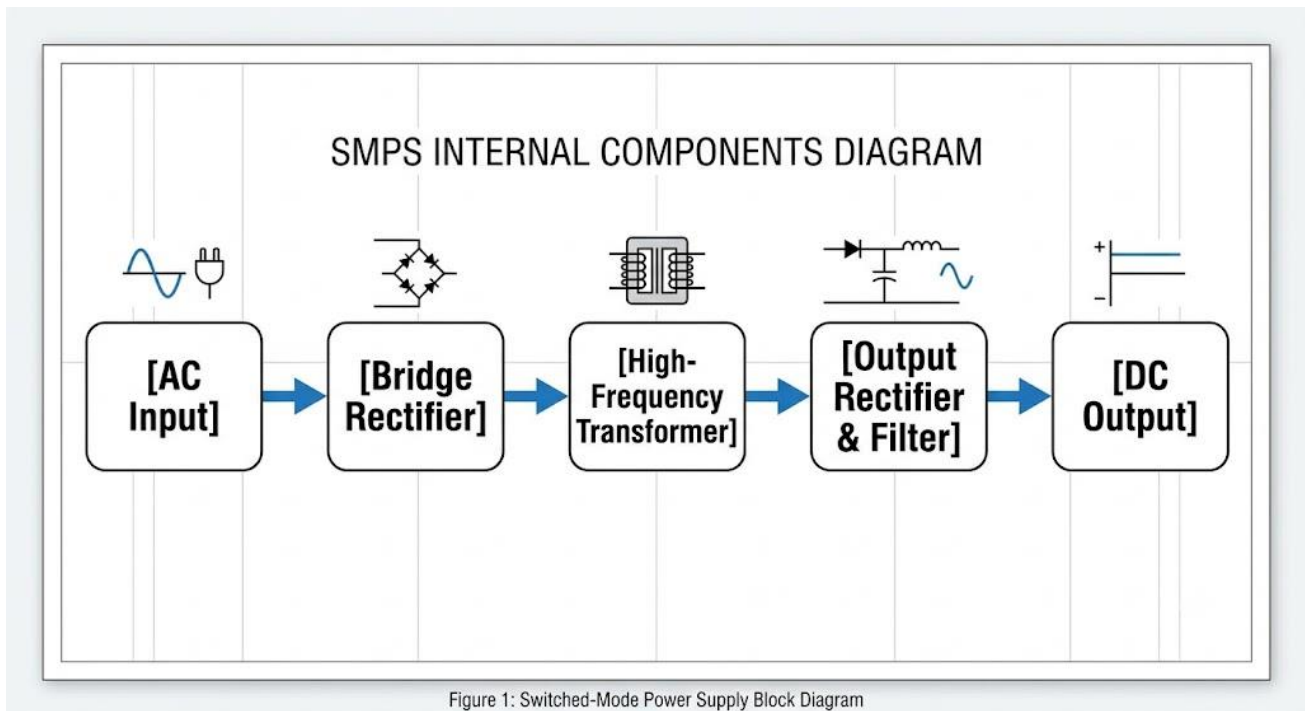


Power Supply Systems Theory

Task 1: SMPS & UPS Devices

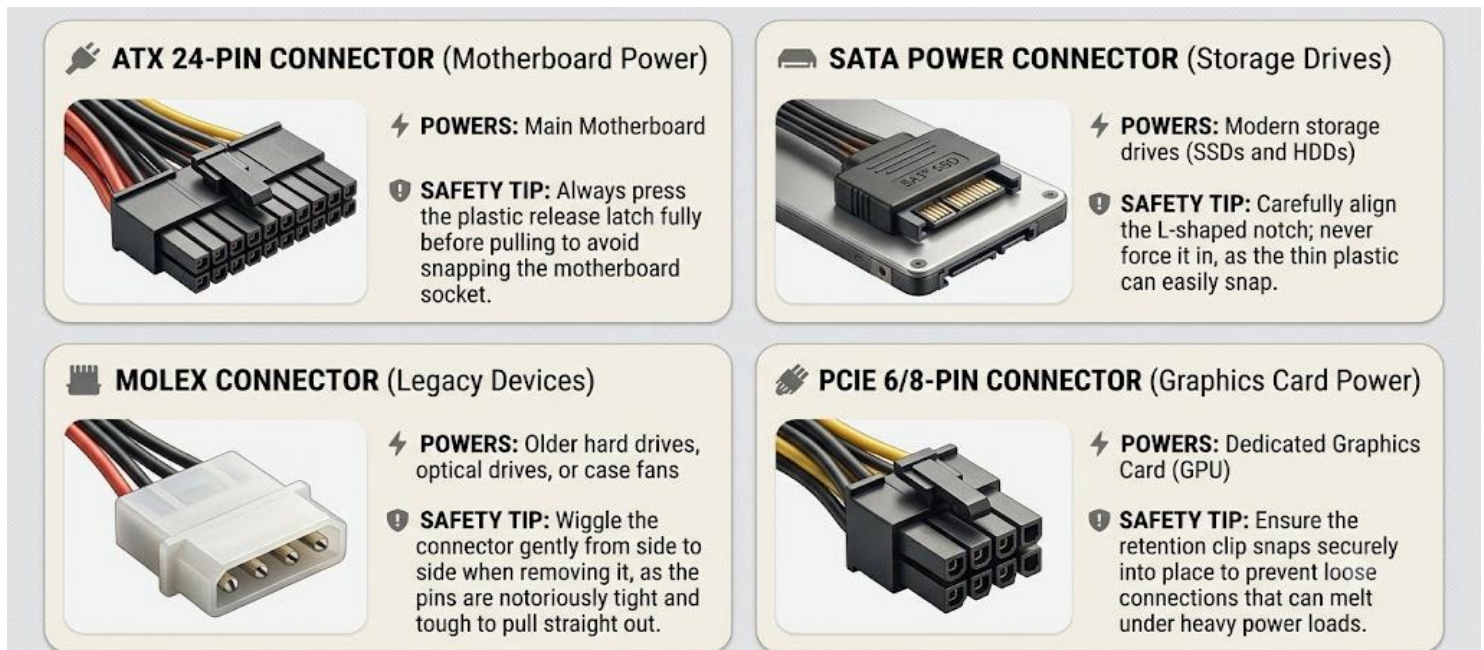
- **SMPS (Switched Mode Power Supply) Devices:**
 - **Desktop Computer:** Needs stable, multiple DC voltages converted efficiently from AC wall power.
 - **Smartphone Charger:** Requires a compact, lightweight way to convert AC to 5V DC power.
 - **LED Television:** Uses it for efficient, low-heat power regulation for internal electronics.
- **UPS (Uninterruptible Power Supply) Devices:**
 - **Desktop Computer:** Requires immediate battery backup to prevent data loss or hardware damage during power outages.
 - **Wi-Fi Router:** Keeps the internet running seamlessly during brief electrical blackouts.

Task 2: SMPS Internal Components Diagram



[AC Input] → [Bridge Rectifier] → [High-Frequency Transformer] → [Output Rectifier & Filter] → [DC Output]

Task 3: Power Connectors



- **ATX 24-pin Connector:** Powers the main motherboard.
 - **Safety Tip:** Always press the plastic release latch fully before pulling to avoid snapping the motherboard socket.
- **SATA Power Connector:** Powers modern storage drives (SSDs and HDDs).
 - **Safety Tip:** Carefully align the L-shaped notch; never force it in, as the thin plastic can easily snap.
- **Molex Connector:** Powers older hard drives, optical drives, or case fans.
 - **Safety Tip:** Wiggle the connector gently from side to side when removing it, as the pins are notoriously tight and tough to pull straight out.
- **PCIe 6/8-pin Connector:** Powers the dedicated Graphics Card (GPU).
 - **Safety Tip:** Ensure the retention clip snaps securely into place to prevent loose connections that can melt under heavy power loads.

Task 4: Gaming PC Connection Steps

1. Mount the SMPS into the computer case and secure it with four screws.
2. Connect the thick **ATX 24-pin cable** to the right side of the motherboard.
3. Connect the **4-pin or 8-pin EPS cable** to the top-left of the motherboard to power the CPU.
4. Plug the **PCIe 6/8-pin cable(s)** into the top or side of the graphics card.
5. Attach the flat **SATA power cables** to the back of your storage drives.